THE SPACES BETWEEN GESTURE, SOUND AND IMAGE

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ABSTRACT

Improvised audiovisual performance is becoming an increasingly rich field of creative practice, fuelled by the image-centricity of contemporary performance culture and increasingly affordable and powerful technology for live digital image projection. However, the domains of audio and visual creativity frequently remain separate. Both theoretical discourse and practical training tend to focus on one domain in exclusion to (or at least with significant priority over) the other.

For example, contemporary literature on new interfaces for musical expression are frequently centred on aspects of design and expressivity within the framework of music performance alone, and discussions on interactive and graphic scores for contemporary music performance focus the affordances of the visual score as a medium. Meanwhile there are limited frames of reference for examining improvised audiovisual performance as an integrated practice.

In this analysis, we focus on integrating the concept of gestural surrogacy from the spectromorphological tradition with concepts of multi-sensory feature binding. To support this investigation, we utilise case-studies of three distinct approaches to audiovisual performance practice, using a combination of motiongram and spectrogram analysis along with artistic comment from the authors of each piece.

1. INTRODUCTION

In examining three improvised audiovisual works, we look at the intrinsic aspects of each authors' process in terms of the individual compositional and performative dialectic of intention and result, including sources of inspiration, creative processes and expressivity. These reflections are necessarily subjective and may offer little to others in terms of directly translatable understanding or re-usable process.

To complement this subjective approach, we also examine the morphological qualities of both the audio and visual elements of each performance, in effect looking at the 'trace' of each audiovisual performance.

Visualisations of this kind are not unprecedented in the realm of analysing complex performances: Borgo (2005) uses fractal correlation as a means of examining the features of improvised musical performances by the like of Evan Parker and others, while Jensenius (2010) and Francoise (2012) have developed systems for analysing physical gestures and sound together.

A multi-sensory approach to understanding visual music need not treat the visual trace as reductive. Rather, it can point to an underlying unification of perception within the body. Key to this perspective is evidence for a close coupling between the cognitive processes for movement and perception. Leman (2008, p. 77-102) provides extensive discussion of the evidence, including the behavioural observation of infants' innate ability to perceive gestures and replicate them, and the neurobiological observation that some of the same neurons which are fired to create a gesture also fire when the subject observes another performing the same action. These are the so-called mirror neurons.

The tight coupling of movement and perception at a cognitive level gives rise to the idea that just as our movements arise from intentions (simulation of the movement), so perceptions of the external world map back to intentions because of the trace left by the shared cognitive processes. This action-oriented ontology suggests that even at the social level, the actions of others are understood in terms our own intentions, i.e. our own simulated actions. The ``moving sonic forms'' of music are likewise attributed with intentionality because of the coupling of perception and movement. Arguably, because human beings develop their own action-oriented ontology by virtue of a relatively common physiology, semantic communication is possible through music, even without a common culture (Leman 2008, p. 92).

By utilising analytical tools which draw directly on the qualities of both sound and motion, our goal is to support a discussion of the intersubjectivity of abstraction and embodiment that manifests in visual music, particularly looking at the idea of gestural surrogacy. While there are many other frameworks for analysing movement and visual aesthetics, for visual music it seems most appropriate to draw upon the existing analytical

tradition of acousmatic music, particularly Dennis Smalley's work on spectromorphology (Smalley 1986, 1997). Smalley's typology is laden with physical metaphors and is sufficiently general so as to be applicable to human movement, and by extension, qualities of visual perception in general, as much as it is to spectrally moving forms in the auditory domain.

At the root level, Smalley considers sound in terms of motions, space, structural functions and behaviours. Motions include both the motion of sounding objects, including such categories as ascent, descent, oscillation, rotation, dilation, contraction, convergence, and divergence. He also suggests several characteristic motions including floating, drifting, rising, flowing, pushing or dragging. One can easily imagine these categories being applied to human movement, at either the individual or group level, and by extension the visual morphologies of audiovisual artworks.

When we consider the gestural relationship between sound and vision in visual music, it is useful to look at the way in which Smalley links spectromophological forms of sound art to the morphology of human movement (Smalley 1997), an insight which is backed by the work of Leman. The key idea here is that we make sense of what we see and hear by drawing upon our experience as embodied beings, even though in visual music the experience to which we attend is generally abstracted away from anything immediately recognisable. From a compositional perspective, Smalley identifies several levels of *gestural surrogacy*, in terms of degrees of abstraction away from both the source material the gestural archetype:

- primal gesture: basic proprioceptive gestural awareness, not linked to music making
- first order: recognisable audiovisual material subject to recognisable gestural *play* without *instrumentalisation*
- second order: traditional instrumental musical performance
- third order: where a gesture is inferred or imagined in the music, but both the source material and the specific gesture are uncertain.
- remote: where ``*source and cause become unknown and unknowable as any human action behind the sound disappears*'', but ... ``*some vestiges of gesture might still remain*'', *revealed by ``those characteristics of effort and resistance perceived in the trajectory of gesture*''.

Building on an awareness of the primal gestural level, first order surrogacy provides simple, immediate accessibility for the interactor, while sustained engagement is generated by providing elements which operate at the higher levels of gestural surrogacy. The aesthesic process is further complicated by the fact that the relationship between audibly perceptible gestures and visually perceptible gestures is not necessarily direct, but exists on a spectrum. In addition, such gestures may not follow the normal rules of physics in the material world (and in the context of an art work, it would probably be quite boring if they did).

A simple example of such gestural mapping is that of ventriloquism: the puppet's mouth movements are linked with audible speech events and although we know that puppet's don't speak, perceptually we are still convinced that the sound is coming from its mouth. Despite the novelty of the effect, the congruence of audiovisual stimuli result in what O'Callahan (2017, p. 162) argues for as perceptual feature-binding: a multi-sensory perception of the same event which may have distinct features such red and rough or bright and loud, but are nonetheless a unified percept. In Smalley's terms, the ventriloquist's performance would contain elements of first order gestural surrogacy.

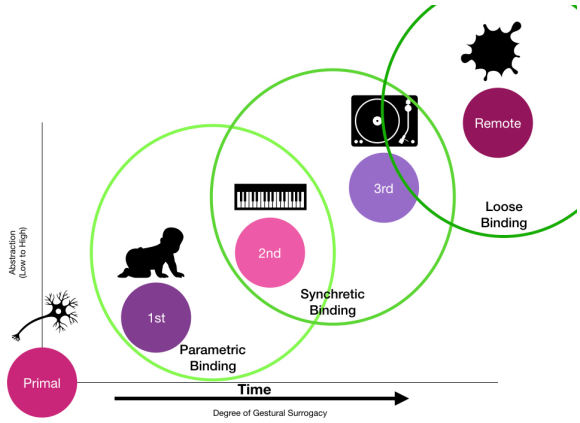


Figure 1. Gestural surrogacy and perceptual feature binding.

Figure 1 shows a mapping between Smalley's concepts of gestural surrogacy, and concepts of perceptual feature binding. Lower-order gestural surrogacy may manifest in visual music through direct parametric mapping between sound and visual elements. For example, the amplitude of a percussive sound being directly linked to the size of a visual element such that repeated patterns of an underlying gesture may result in an experience of auditory and visual pulsing. In this way, the size of the visual object and the amplitude of the sonic object may be perceived to be the result of the same gesture of `striking', even though in our experience of the material world, while percussive instruments may in fact make a louder noise the harder we hit them, visual objects do not routinely become bigger when we hit them.

The same approach could be taken to linking sound elements which may have a floating or gliding quality with, for example, visual elements that also have the same quality of movement. Such a relationship may be parametric, as might be observed via visualisation of audio as a waveform, where a gently undulating tone may have a similar quality to its shape. Or the relationship may be formed through synchresis, that is, simply if two elements appear within the relevant perceptual stream at

the same time, without having any explicit parametric mapping at a compositional level.

It can be difficult to distinguish between parametric mapping and synchresis, as the mapping is typically hidden and only inferred (we imagine the ventriloquist is controlling the puppet, but we typically don't see it explicitly, we only see the puppet's movements synchronising with audio events). This is the paradoxical challenge of a well-realised mapping process: when the mapping is seamless, the audience assumption can be that events in one sense modality have just been composed to synchronise with events in the other, rather than having one arise from the other. Live performance can help reduce this effect by demonstrating that multi-sensory events are arising from a common gestural source in the moment, rather than being pre-composed.

Which brings us to the point that it is important to distinguish between the degree of unification within an audiovisual event (tight versus loose feature-binding) and the degree of gestural surrogacy. If we consider gesture, whether it be in performance or on-screen or auditory, to be a recognisable pattern of experience, then gestures can be thought of as arising from the features of an event as it unfolds. Certainly tightly bound features, recognisable elements within an event, which consistently co-occur or share the same qualities, may be perceived as being the result of first or second order gestures. Such an inference arises because of embodied cognition: our experience of physical movement informs our experience of surrogate forms of movement in other `bodies'.

Loosely bound features bring uncertainty into the experience of an event: is there a common gesture, a common cause? Are these phenomena even linked? Within such uncertainty, it is harder to recognise the gesture behind them, however that does not mean it is not there. At higher degrees of gestural surrogacy, we may experience less overt, conscious recognition, receiving only hints or vestiges of an underlying pattern. The original gesture may be disrupted or distorted in some way, through interruption, amalgamation or elongation, and yet still evoke a response.

To draw together the various threads of this background discussion, we suggest that within visual music, the spectrum of multi-sensory gestural associations arise from, and abstract our own embodied cognitive processes, and that through examining both the creative intent of the composer/performer and the various gestural traces, however we might obtain them, a deeper understanding emerges. Within this framework of understanding, we present reflections on three audiovisual performances. These are examined as an engagement with the intersection of multiple perceptual modes which operate as a space for collaborative art-making beyond the confines of textuality and symbolic representation. Our hope is that as a result, the reader is inspired to engage more deeply in their own experience of the form.

2. WIND SOUND BREATH

Wind Sound Breath is a visual music[1] work by Brigid Burke, performed in this instance by Nunique Quartet (Brigid Burke, Steve Falk, Megan Kenny and Charles MacInnes). This piece utilises densely layered and heavily processed visual material incorporating elements of video footage and graphic elements. In this piece, the visual material is fixed and performances are largely improvised, with certain recognisable elements used to mark sections of the performance.

The piece incorporates live and pre-recorded interactive audio electronics, live visual footage and synchronized video for live performance. Thematically, the piece draws inspiration from Yarra Bend Park, the natural bush land near Melbourne Polytechnic. Wind-Sound-Breath explores connections with the present histories within the campus, Burke's own musical influences and the importance of the Yarra Bend as an ecological site. In the performance analysed here, live instruments included clarinet, percussion, flute, trombone, and live electronics. An overview is given in Figure 2.

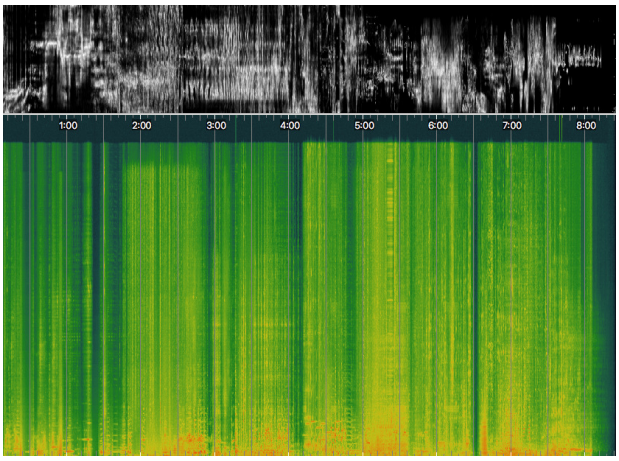


Figure 2. *Wind Sound Breath* Overview

Figure 3 shows detail of an excerpt between 1:46 and 2:17. Sparse clarinet and flute notes enter in the first 10 seconds, distributed against a background of field recordings of ocean waves and a wood fire. Visually, dense root-like patterns fade in and out with slow pulses of colour, modulated by a rippling visual overlay. The motiongram shows the periodic pulsing of this segment and spectrogram show a number of loose correlations between the visual pulse and bands of broadband noise, in this case arising from modulations of the volume of the ocean and wood fire recordings.

[1] In this context, visual music is considered to be a work in which both audio material and on-screen visual material are integrated elements of the total work, allowing for one or both modalities to be improvised or fixed.

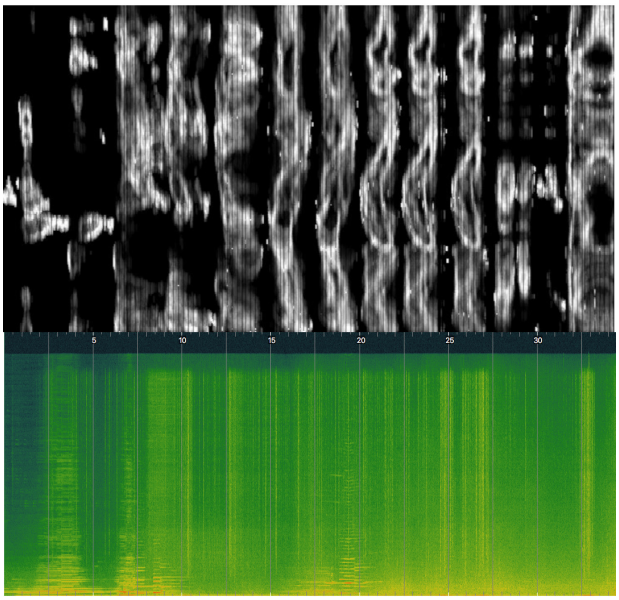


Figure 3. *Wind Sound Breath* excerpt 1.

Figure 4 shows detail of an excerpt between 3:50 and 4:10. This much more active section features lively figures of trombone, clarinet and flute over timpani, with elements of reversed tape material. Visually the underlying root-like pattern persists, layered with footage of monochromatic lapping water and highly colourized, horizontally scrolling footage of a cityscape. Approximately halfway through the excerpt, the colourized city footage begins to shrink behind the monochromatic water footage. A corresponding drop in activity in the motiongram can be seen from this point. Likewise, activity in the audio performance drops away, with less volume and density compared to the first half.

From Burke's perspective as the composer, the aesthetic focus of the work is on exploring relationships between composition, improvisation, visual impact and listening. While pre-composed visual elements are utilised, these are used in improvised video projection performance alongside the improvised audio performance. Audio performers are free to interpret the visual score as it unfolds, and the visual material itself is modified as part of the performance.

In speaking about her process, Brigid Burke explains that concepts for a piece often start with a visual structure in mind, which then drives her to seek out sounds which match that internal visualisation. In terms of actual material, sound always comes first, after which visual elements are created or collected to fit with the audio material. This kind of round trip between an internally visualised concept, sonically realised and followed by a visual realisation, keeps the various elements tightly bound. Burke notes that she's found if she starts with creating visual material first, the piece tends to lack coherence compared to pieces which have audio as the material foundation.

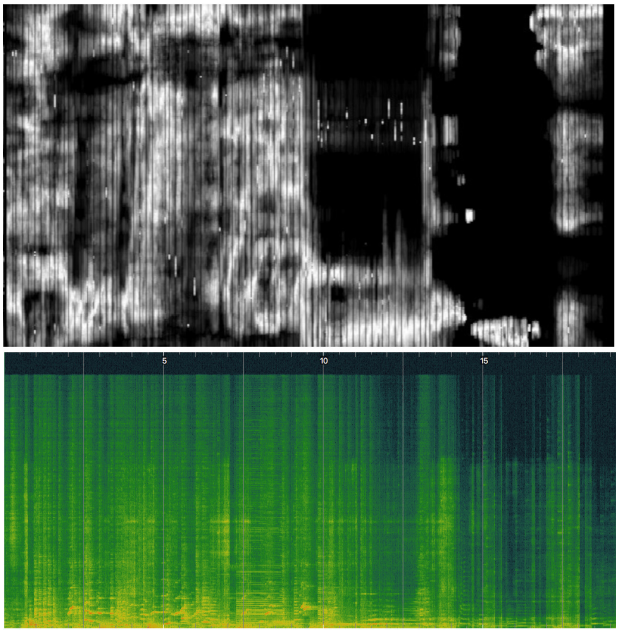


Figure 4. *Wind Sound Breath* excerpt 2.

For live performance, Burke is frequently manipulating one or more elements of either the audio or visual material in addition to live instrumental performance. Live clarinet performance is combined with either live video mixing or live manipulation of synthetic audio elements.

In ensemble performances, the visual element acts in terms of setting the dynamics of instrumental improvisation. This can be seen in the analysis above, with various dynamic changes synchronising across both audio and visual elements. However, the relationship is not just one way, from visual score to audio performance. For Burke, the relationship with other players in the ensemble are of equal weight in terms of shaping the dynamics of her playing and manipulation of either video or audio elements. Thus, while not directly parametric, there is a degree of feedback between audio performance and visual performance which binds the work into a multi-sensory whole.

3. SHIVER AND SPINE

Roger Alsop's piece *Shiver and Spine*, for solo guitar and responsive visuals, demonstrates another approach to audiovisual improvisation: one in which visual material arises from the sonic performance, as opposed to improvisations which respond to fixed visual media. A dynamic visual element, depicting the apparent victor of recent internal political struggles within the Australian Federal Government, leaps on the screen in response to distorted gestures from an electric guitar. Figure 5 shows a still from the performance, while Figure 6 provides an analytical overview of the work.



Figure 5. *Shiver and Spine* still.

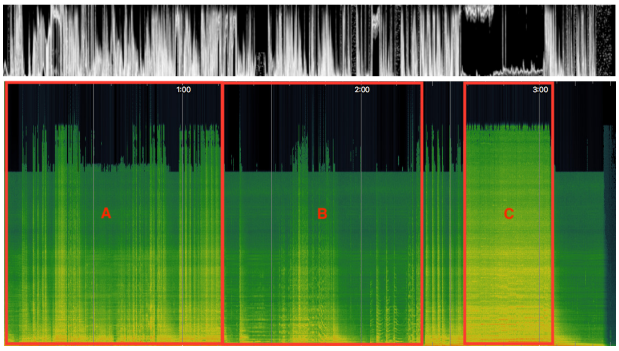


Figure 6. Analysis of *Shiver and Spine*.

In the overview, three episodes are marked. Episode A (from 0 to 1:12), introduces the elements of the piece, with video elements distorting in terms of shape, luminance and colour in response to distorted chords and textural scratching from the guitar. During this section, audio and video appear to be tightly synchronised. Episode B (from 1:12 to 2:25) is comparatively sparse, with occasional struck chords left to decay before more gentle tapping of strings is used to re-introduce movement. Video elements continue to be active during the decay of each chord, calling into question exactly what drives the motion - is it frequency modulation rather than attack, or is there a less direct relationship than had previously been suggested by the performance during episode A?

A short section of stronger struck chords introduces episode C (from 2:35 to 3:05), which shows a distinct drop in visual activity in the motiongram with a corresponding sustained block of broadband audio activity in the spectrogram, generated by fierce strumming by Alsop. The responsive visual material is literally saturated by this block of audio input, filling the screen with a red field of colour - an extreme close up of the piece's subject.

It is conceivable that without the image of the performer and the guitar in the room, there would be no way to determine if the audio or the visual elements were dominant. In fact, only through a tacit understanding of the technical processes, coupled with the tightly synchronous performance, would the audience be able to determine that the visuals are responding to the audio and not that the performer is responding to the visuals. Like Burke's *Wind Sound Breath*, *Shiver and Spine* does set up a feedback loop between the responsive visuals and the audio performer: at a gestural level, visual response invites audio exploration, and perhaps at the broader discursive level, the sheer repetition of the subject's image, pushes the audio performer into a moment of sonic rage such as we see in episode C.

Alsop frequently creates works which utilise audio-responsive visual elements and does so in a way which enfolds the performer and the audience within a recursive loop of stimulus and response. He observes that many interactive audiovisual artworks function to augment human performance, and in this way work as an instrument rather than an interlocutor. In many of his works, rather than create work that unifies or coheres disparate elements, Alsop seeks to interpret human gesture of performance as a different artwork, to be viewed without and separate from the human element involved in its creation. In this regard, he considers it important that when interacting with systems for audiovisual improvisation, performers experience their own actions are represented in an unfamiliar way, so that they do not revert to familiar styles or clichés similar to those they may use when interacting with another performers (dancers, musicians).

4. APOPHENIC TRANSMISSION

Apophenic Transmission (i) is part of a series of live performance works which interrogates the human tendency to spontaneously perceive connections and meaning in unrelated phenomena. In this work the performer is challenged to seek occasions of apophenia through the use of a somewhat opaque and chaotic control interface, in much the same way as one might attempt to tune a radio or a television set to a dead channel for the purpose of encountering electronic voice phenomena. Both the audio and the visual elements are purely generative, with the audio arising from a software simulation of the Synthi AKS, and the visuals arising from software simulation of an analog video synthesier.

Part of the purpose of this piece is to call into question the parametric mapping approach to visual music, which Pedersen frequently uses. While the physical performance controls used for both the generative audio and video elements overlap, the performance framework for *Apophenic Transmission (i)* deliberately confuses this control interface, so that both performer and audience have cause to question whether there is any direct relationship between what is heard and what is seen. As Garro (2012) suggests. ``*visual music is more than a mapping exercise; the absence of a relationship is itself a relationship*".

Figure 7 gives the spectrogram and motiongram analysis for the whole piece. Compared to the audiovisual analysis of the preceding works, it is no surprise that observable correspondences between sound and vision are hard to extract.

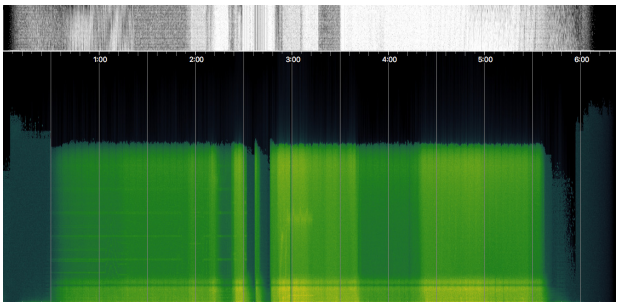


Figure 7. *Apophenic Transmission (i)* overview.

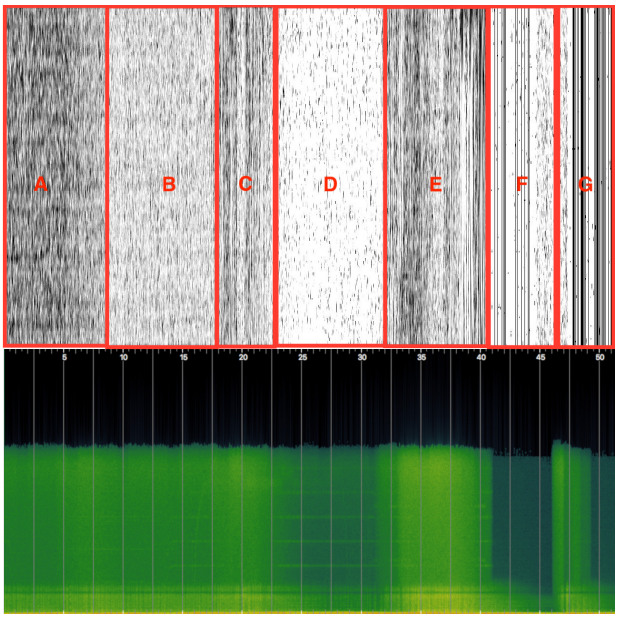


Figure 8. *Apophenic Transmission (i)* excerpt 1.

Looking more closely at an excerpt from between 1:51 and 2:42, shown in Figure 8, we can start to identify features which may trace a common underlying gesture. The motiongram shows several distinct episodes, labelled A - G. In the spectrogram we can see corresponding changes in each of these episodes, although some are more subtle than others.

Comparing screenshots from episodes B and D, as shown in Figures 9 and 10, there is a shift from visual static to a more structured pattern. In episode D, the width of the visual bands pulse dynamically. There is broadband noise in episode B, which drops away in episode D to reveal beating sine tones, which can be seen as horizontal bands across the spectrogram. These same bands are also faintly apparent in the spectrogram of episode B, but the broadband noise masks them. Episode E exhibits the same qualities as B, and episode F exhibits the same qualities as episode D. Thus, the piece starts to establish a gestural link between the presence of audio and visual noise, which masks other underlying patterns.

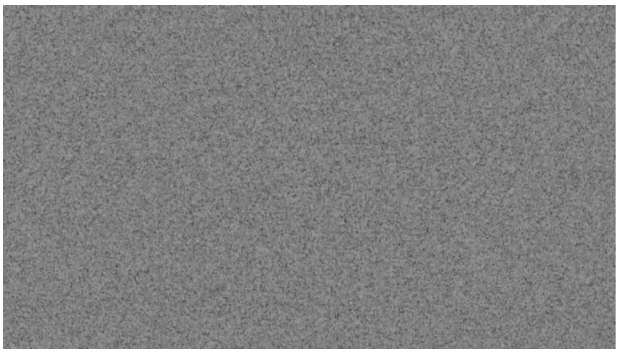


Figure 9. *Apophenic Transmission (i)* still 1.

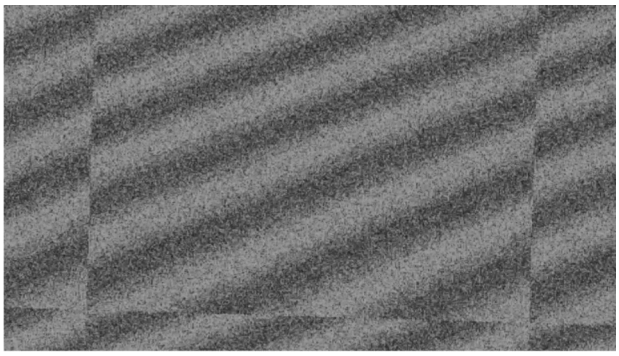


Figure 10. *Apophenic Transmission (i)* still 2.

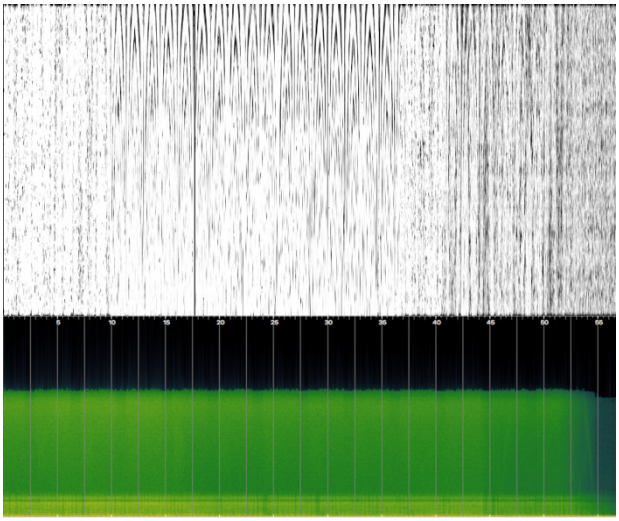


Figure 21. *Apophenic Transmission (i)* excerpt 2.

Figure 11 shows a segment close to the end of piece. High frequency noise still sits across this section, but it decreases in energy across the episode, becoming weaker and lower in frequency between 50 seconds and the end of the clip. Visually we can observe a distinct oscillating pattern across the top of the motiongram between 10-40 secs, which corresponds to a pattern of horizontal bands expanding and contracting in a general downward rolling bounce, typical of the classic bouncing ball motion of modulated low frequency oscillator. The audio spectrogram shows periodic punctuations in the high frequency noise band, which correspond to the lowest point of the visual oscillation (when the horizontal bands are at their widest). The strength of this pattern gradually

declines from 35 seconds onward, with lower contrast visually, and a corresponding reduction in high frequency noise content.

Pedersen notes that for him, performances start *"in the body"*. His internal feeling is the generative core of each performance, and physical expression of that emotion is typically an essential component. His aim is seeking sounds, or rather transformations of sounds, which match the expressive quality of that internal feeling. In this sense, sound becomes the vehicle for expressing the internal (emotional) state. The range of expression could be very dynamic or very subtle, and the interfaces which support the appropriate range of gestures are a key consideration.

In this regard, a performative aspect to each piece is important, and correlations between gesture and sonic effect are highly desirable. For dynamic pieces, interfaces which allow a range of performative gestures, such as Kinect motion sensors, are preferred. Subtle performances are more easily accommodated via traditional knob and slider interfaces. Each system may still need to be learned in terms of gesture mapping. Difficulty in performance is ok, and often welcomed, as the challenge of various gestures can be part of the desired emotional expression. Likewise, it is sometimes important to not know or understand the mapping. In this case the feeling being sought is one of loss of control / confusion.

This is the case for *Apophenic Transmission (i)*, where the control surface used for performance is mapped into both the audio and visual generation systems, which otherwise are independent from each other. The mapping is deliberately unstable and unrehearsed - leading to accidental, or unintentional correspondences as the performer seeks to understand the affordances of the control surface, both visual and auditory, just as the audience is also seeking to make sense of the system.

5. CONCLUSION

In considering visual music, particularly where there is an element of improvisation, the body is a natural starting point for sense making, since as embodied beings, sound cognition is intimately linked with the way our brains also process movement. Yet as habitual meaning makers, bare experience is not sufficient: there is a human tendency to ascribe intent to what we perceive as gestures, however abstract, and connect those gestures with our own personal history and framework for understanding the world.

When visual music, as a genre, actively resists narrative, which may be considered the realm of cinema, then what remains is affect - the mood or the feeling, that which is present before narrative emerges, or perhaps that which remains after narrative has been taken away. Furthermore, we are specifically interested in the nexus of audiovisual performance as a field of affect, as distinct from that which may arise from just a single channel auditory or visual experience. Rutherford (2003) suggests that *"shape, colour, texture, protrusions and flourishes all reach out and draw us to them in an affective resonance"*, particularly highlighting the work of Michael Taussig on mimesis, quoting:

> *"It is not the mind's eye that reaches out to grasp or grope the image or space before me—it is my embodied self locating, placing myself in the world which I am viewing."* Taussig (2018)

With this in mind, and drawing upon O'Callaghan's (2017) argument that perception is truly multi-sensory, it is perhaps useful to consider visual music as being an integrated space of *"affective resonance"*. Within this mimetic space, it is the quality of the gestures embodied within audiovisual art works, however abstracted they may be, which evoke engagement with the audience.

Considering the examples analysed here, what affective resonances arise?

In *Wind Sound Breath* there is perhaps a sense of playfulness in the performance as a whole as it shifts between delicate, interlocking filigrees and occasions of robust wrestling. *Shiver and Spine* is comparatively much more raw, possibly evoking anger or at least frustration in the urgent, distorted guitar, as both performer and audience alike are taunted by a kind of jack-in-the-box visual element. In contrast, *Apophenic Transmission (i)* evokes a degree of confusion, a tentative, exploratory questioning or doubt about what is being seen and heard. With little that is overt in terms of gesture, the experience is more akin to a half-remembered dream, for both performer and audience.

In reflecting on our experience with hosting audiovisual improvisation events for more than six years, the experience of liveness that arises from the sharing of gestures in the moment is the compelling reason to continue participating, for creators, performers and audience members alike. If, as Taussig suggests, the act of perception is itself also a gesture, then we are all seeking those moments of contact that arise beyond (or beneath) narrative, in the space between gesture, sound and image.

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